

Where you stand in Chapter 2

WRITE THIS CODE ON YOUR PAGE

D2

10 questions, chosen to split what is still uncertain about your skills

HOW TO WRITE YOUR WORK

- Write the code **D2** at the top of every page you photograph.
- Start each problem with its number. One step per line.
- Cross mistakes out with a single line; do not erase.
- Box your final answer.
- If you have not seen a topic yet, write the number and *skip*. Skipping is information too.

1. Simplify $x^2 + 7x + \frac{12}{x+3}$ for $x \neq -3$.

2. Write $|x - 3|$ without absolute value bars, for $x > 3$.

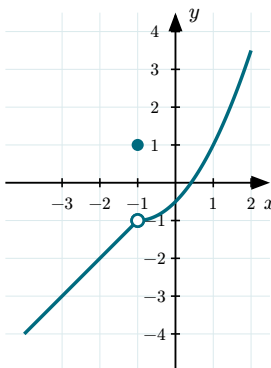
3. Find the exact value of $\tan\left(\frac{7\pi}{6}\right)$.

4. Let $f(x) = \begin{cases} 2x - 1 & \text{if } x \neq -1 \\ 0 & \text{if } x = -1 \end{cases}$. Find $\lim_{x \rightarrow -1} f(x)$.

5. Use the table to estimate $\lim_{x \rightarrow 0} \frac{e^{3x} - 1}{x}$.

x	$\frac{e^{3x} - 1}{x}$
-0.1	2.591818
-0.01	2.955447
-0.001	2.995504
0.001	3.004505
0.01	3.045453
0.1	3.498588

6. Use the graph of $y = f(x)$ to find $\lim_{x \rightarrow -1^-} f(x)$.



7. Let $f(x) = \begin{cases} x^2 - 1 & \text{if } x < -2 \\ x - 1 & \text{if } x \geq -2 \end{cases}$. Find $\lim_{x \rightarrow -2^+} f(x)$.

8. Find all vertical asymptotes of $f(x) = \frac{x^2 - 3x + 2}{x^2 + x - 2}$.

9. Evaluate $\lim_{x \rightarrow -1} \frac{2x^2 + x - 6}{x - 5}$.

10. Find the value of k that makes $f(x) = \begin{cases} kx + 4 & \text{if } x < 2 \\ x^2 + 5 & \text{if } x \geq 2 \end{cases}$ continuous everywhere.